PART 4: SEQUENTIAL LOGIC

REGISTERS

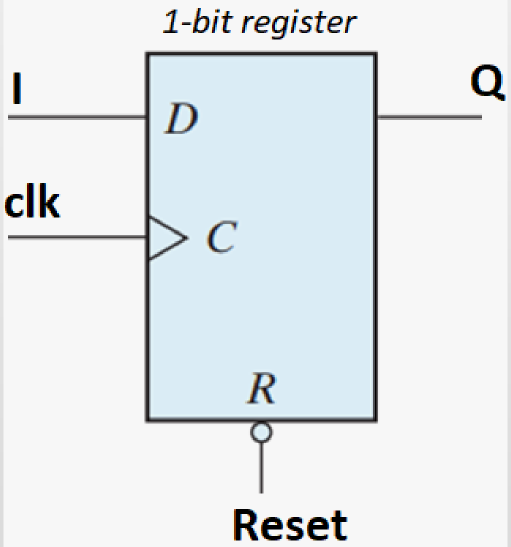
SAAD BAIG – DIGITAL LOGIC DESIGN

REGISTERS

- Registers are a type of sequential logic circuit used primarily for the storage of digital data.

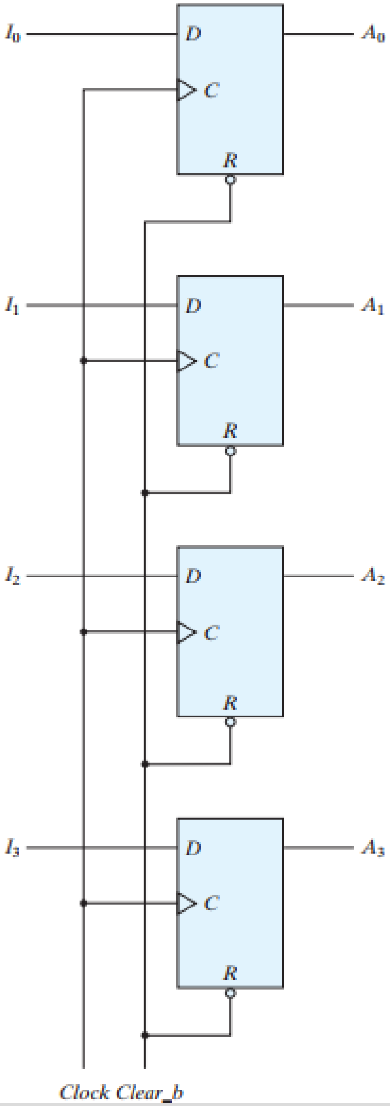
A register consists of a group of flip-flops together with gates that affect their operation.

- The flip-flops hold the binary information, and the gates determine how the information is transferred into the register.
- An *n*-bit register consists of a group of *n* flip-flops capable of storing *n* bits of binary information.

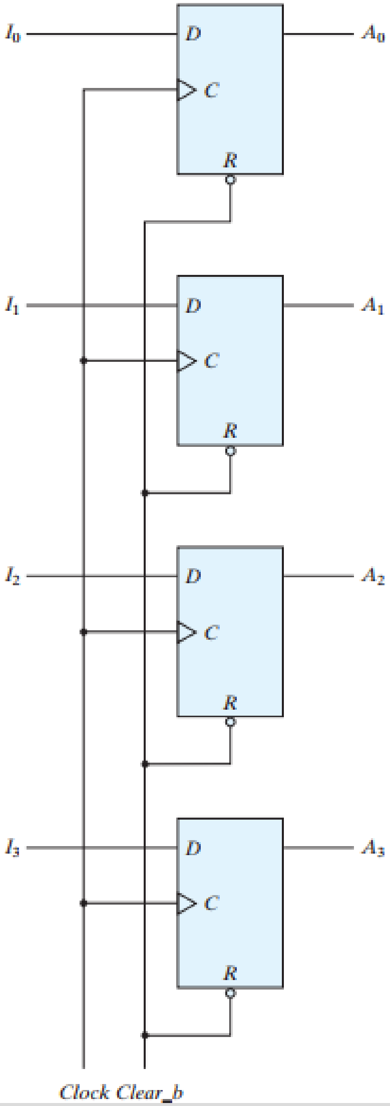


REGISTERS WITH PARALLEL LOAD

- In this configuration, the common clock input triggers all flip-flops on the positive edge of each pulse.
- The values of I_3, I_2, I_1, I_0 immediately before the clock edge determines the values of A_3, A_2, A_1, A_0 after the clock edge.
- The *Clear_b* input is useful for clearing the register to all 0's prior to its clocked operation. It must be maintained at logic 1 (i.e., de-asserted) during normal clocked operation.



- The transfer of new information into a register is referred to as loading or updating the register.
- If all the bits of the register are loaded simultaneously with a common clock pulse, we say that the loading is done in parallel.
- In this configuration, the common clock input triggers all flip-flops on the positive edge of each pulse.



REGISTERS WITH PARALLEL LOAD

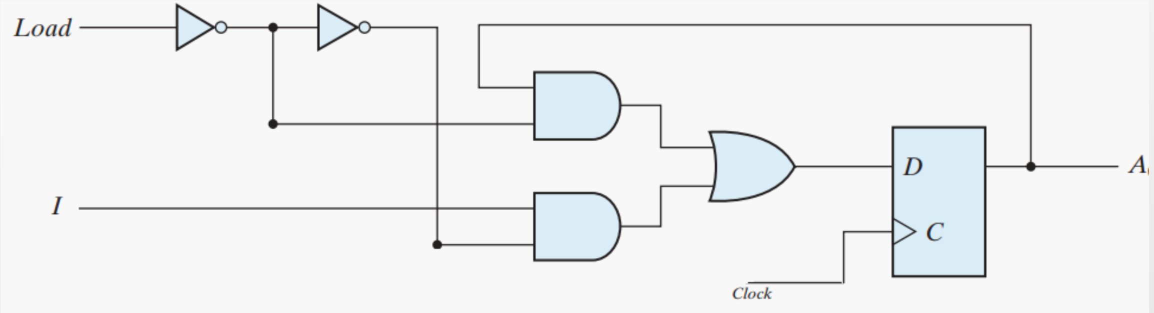
REGISTERS WITH PARALLEL LOAD

- If the contents of the register must be left unchanged, either:
  1. The clock must be inhibited from the circuit, or
  2. The inputs to flip-flops must be held constant.
- In the first case, the data bus driving the register would be unavailable for other traffic.
- In the second case, the clock can be inhibited from reaching the register by controlling the clock input signal with an enabling gate. However, inserting gates into the clock path is ill advised because it means that logic is performed with clock pulses.
- The insertion of logic gates produces uneven propagation delays between the master clock and the inputs of flip-flops.

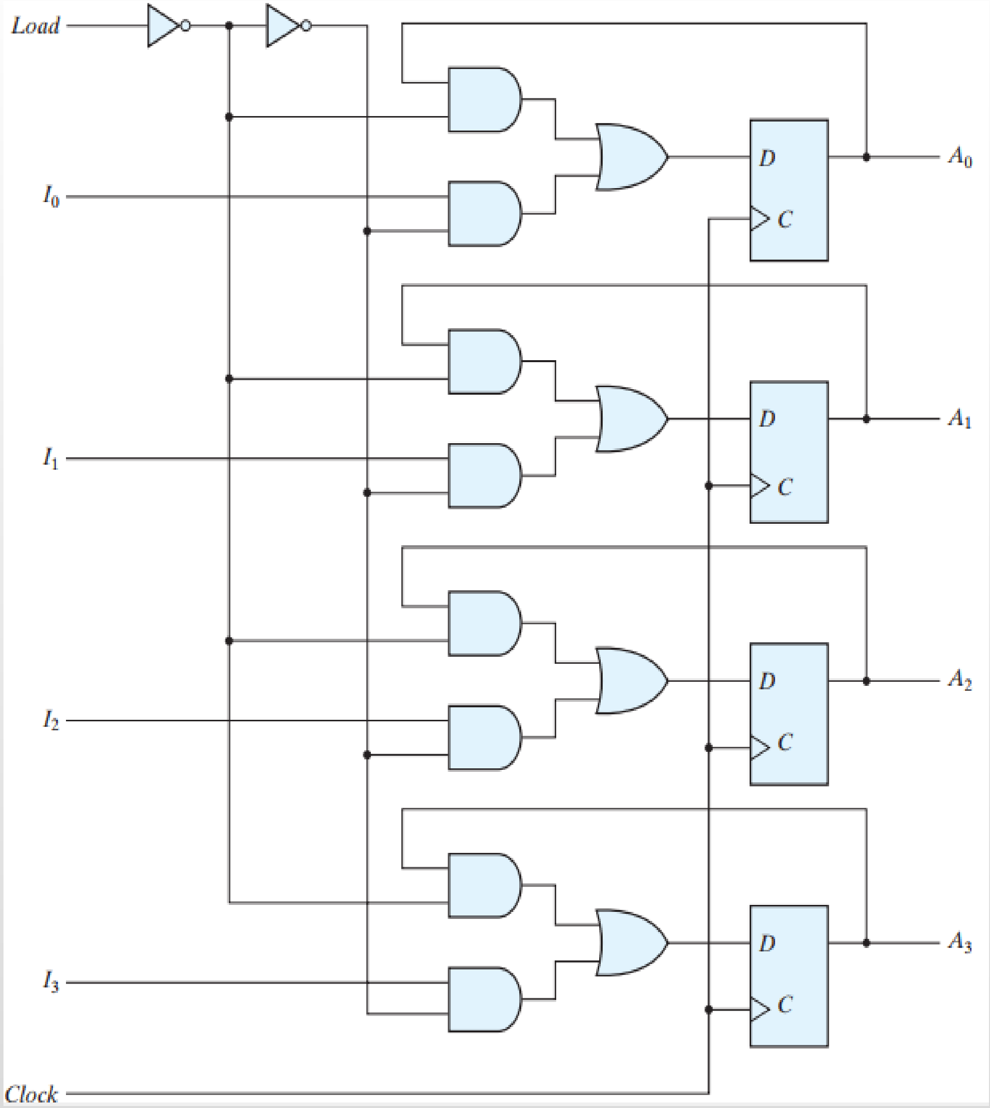
REGISTERS WITH PARALLEL LOAD

- To fully synchronize the system, we must ensure that all clock pulses arrive at the same time anywhere in the system, so that all flip-flops trigger simultaneously.
  - Performing logic with clock pulses inserts variable delays and may cause the system to go out of synchronism.
- For this reason, it is advisable to control the operation of the register with the D inputs, rather than controlling the clock in the C inputs of the flip-flops.
  - This creates the effect of a gated clock, but without affecting the clock path of the circuit.

REGISTERS WITH LOAD CONTROL INPUT



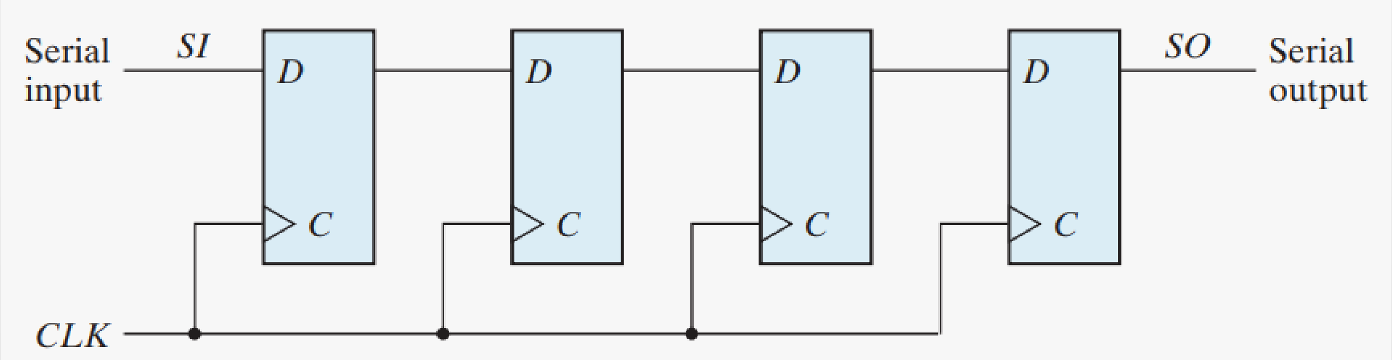
- The additional gates implement a two-channel mux whose output drives the input to the register with either the data bus or the output of the register.
  - When the load input is 1, the data at the external inputs are transferred into the register with the next positive edge of the clock.
  - When the load input is 0, the outputs of the flip-flops are connected to their respective inputs.



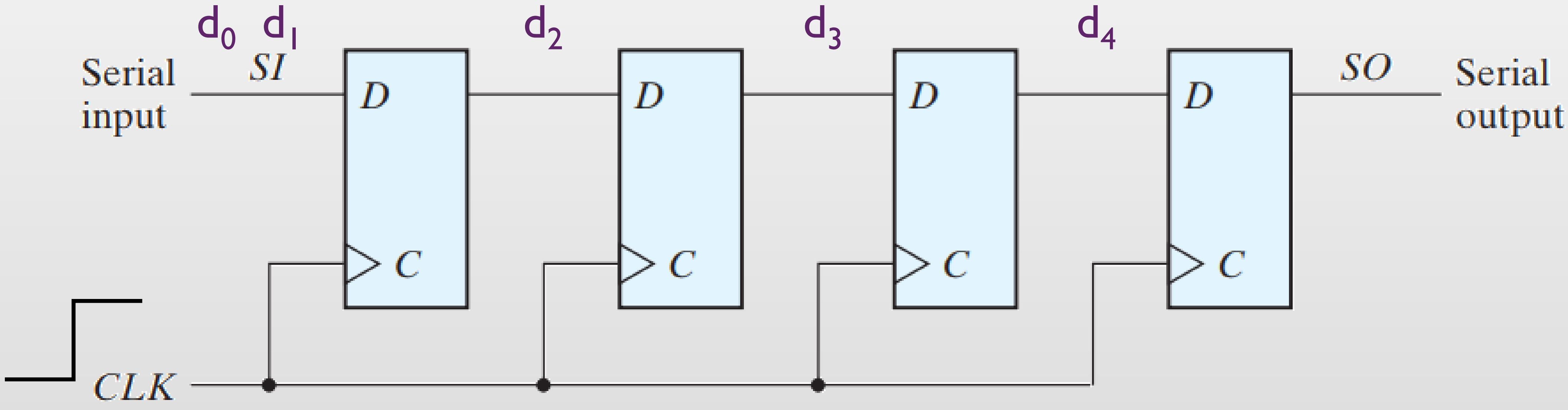
4-BIT DATA STORAGE REGISTER

SHIFT REGISTERS

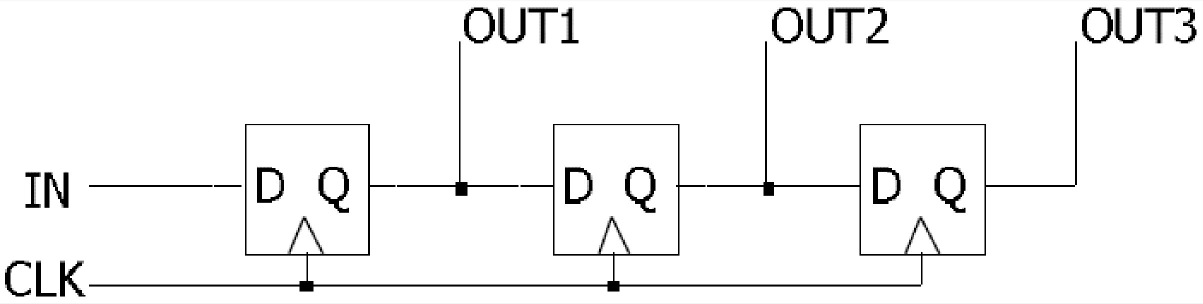
- A register capable of shifting the binary information held in each cell to its neighboring cell, in a selected direction, is called a shift register.
- Chain of flip-flops connected in cascade. All flip-flops receive common clock pulses. Each clock pulse shifts the contents of the register one bit position to the right.



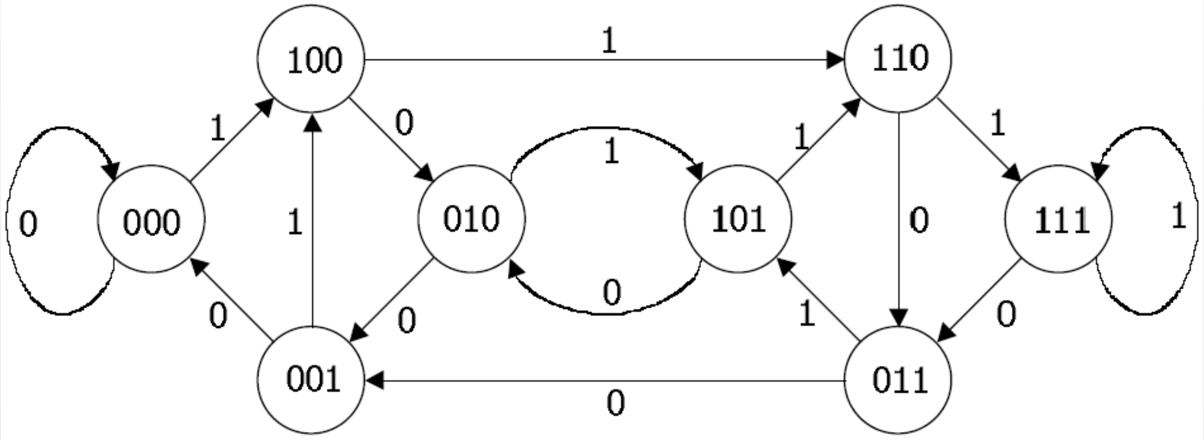
4-BIT SHIFT REGISTER



3-BIT SHIFT REGISTER

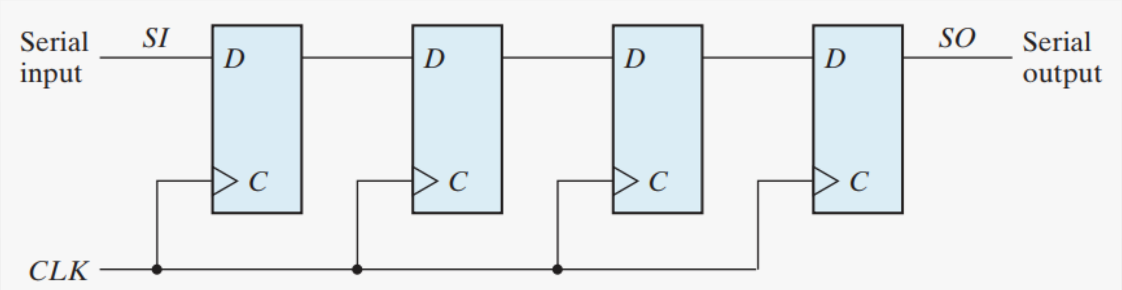


- Any sequential circuit can be represented with a state diagram.



SHIFT REGISTERS

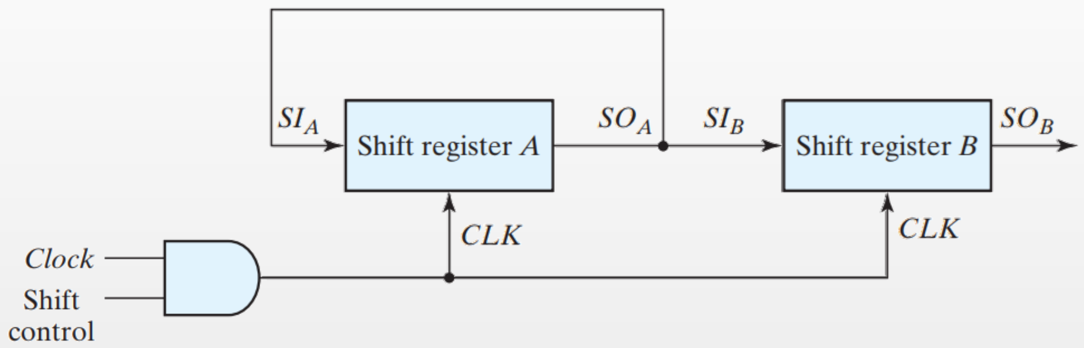
- Does this configuration support a left shift?
- Sometimes it is necessary to control the shift so that it occurs only with certain pulses, but not with others.

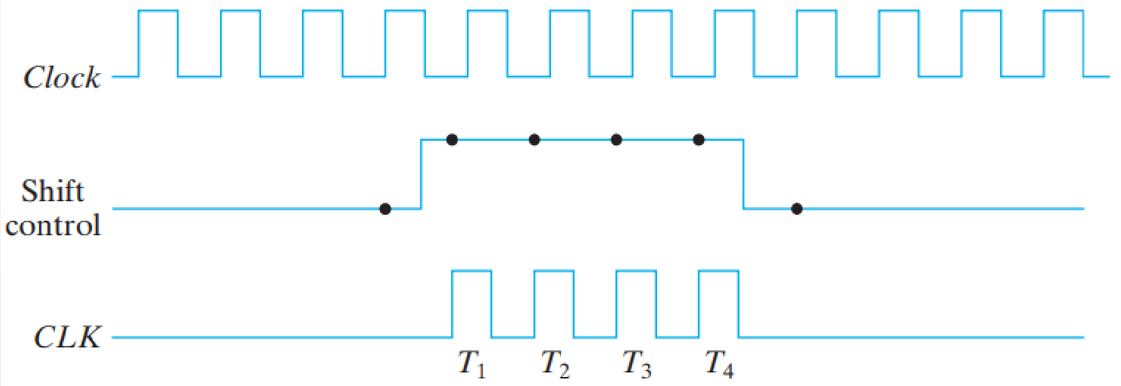


- As with the data register, the clock's signal can be suppressed by gating the clock signal to prevent the register from shifting.
- A preferred alternative in high-speed circuits is recirculating the output of each register cell back through a two-channel mux whose output is connected to the input of the cell.

SERIAL TRANSFER

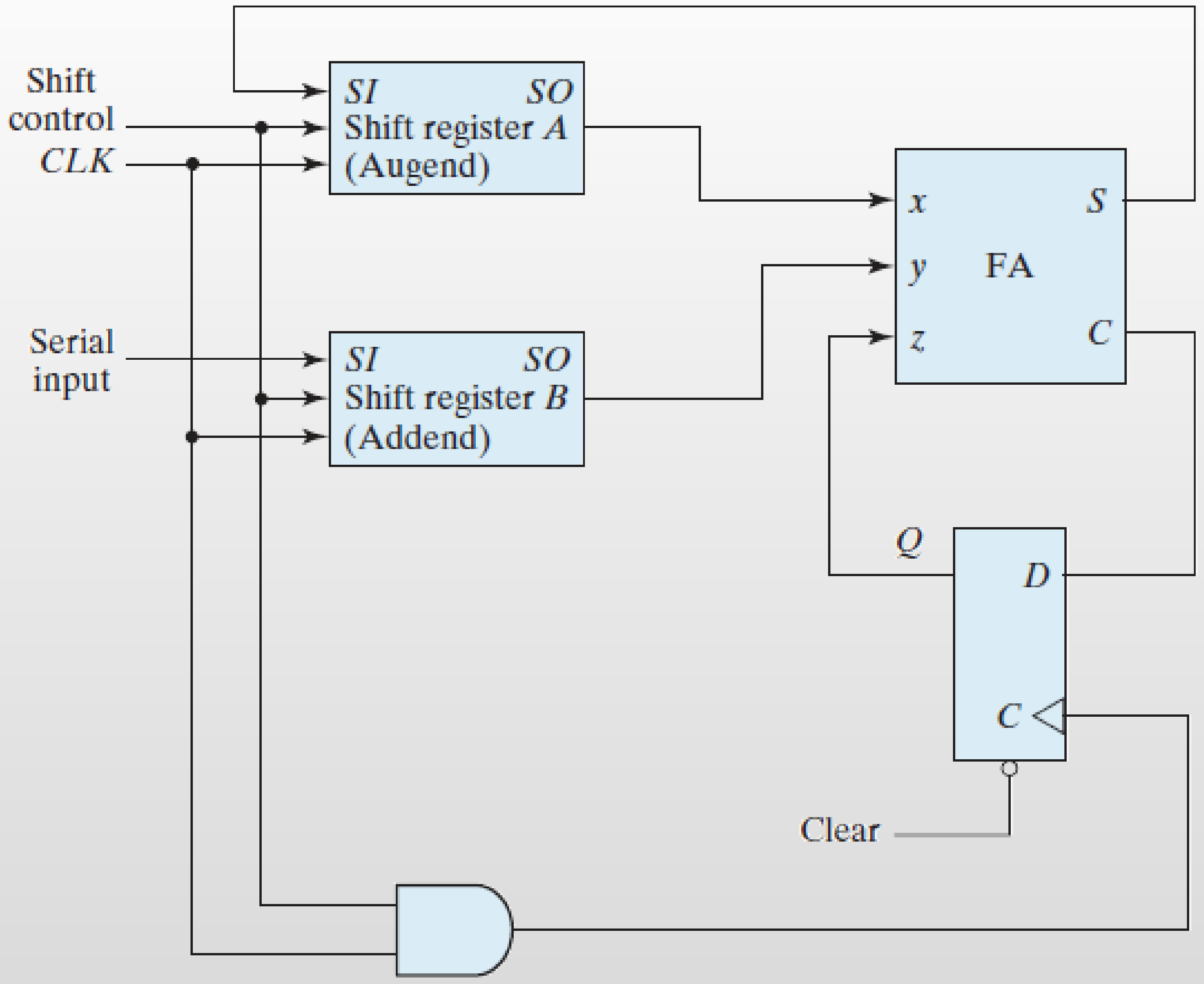
- The data path of a digital system is said to operate in serial mode when information is transferred and manipulated one bit at a time.
- Information is transferred one bit at a time by shifting the bits out of the source register and into the destination register.
- This type of transfer is in contrast to parallel transfer, whereby all the bits of the register are transferred at the same time.



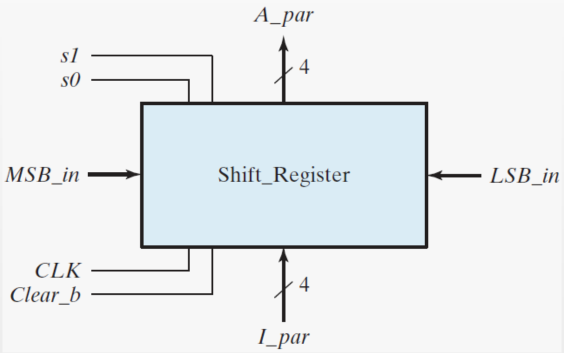


Timing Pulse	Shift Register *A*				Shift Register *B*			
Initial value	1	0	1	1	0	0	1	0
After T_1	1	1	0	1	1	0	0	1
After T_2	1	1	1	0	1	1	0	0
After T_3	0	1	1	1	0	1	1	0
After T_4	1	0	1	1	1	0	1	1

SERIAL TRANSFER

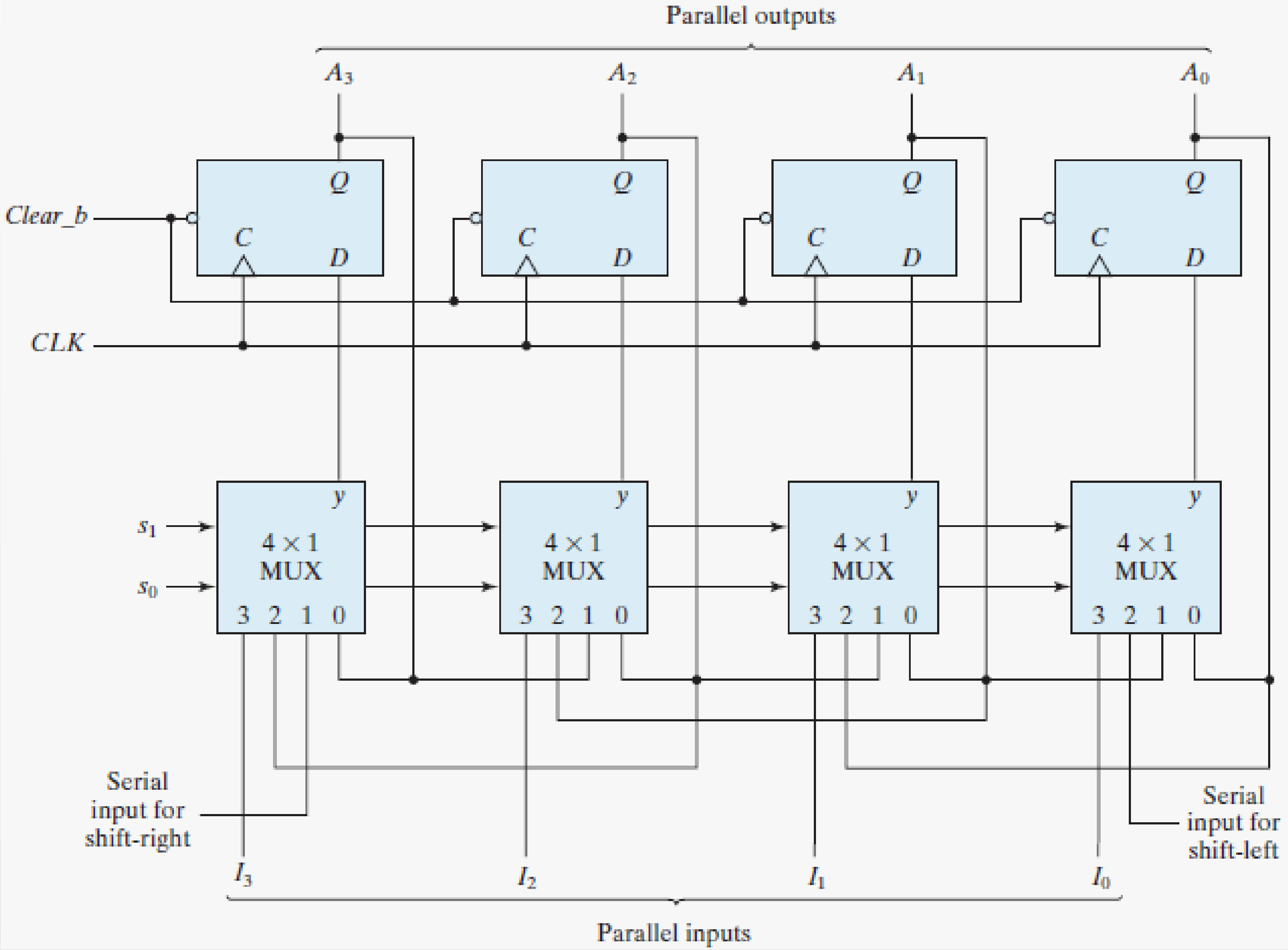


SERIAL ADDITION



Function Table for the Register

Mode Control		
s_1	s_0	**Register Operation**
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load



UNIVERSAL SHIFT REGISTER